

Winter Water Quality Survey

Alder Creek Field Station · Reach 4 · November 2025

Summary

Five monitoring sites were visited over three days in November. Dissolved oxygen was high at every site, as it usually is at this temperature, and pH was steady across the reach. Conductivity was unremarkable and is recorded here as the baseline the spring survey should be read against.

Method

Readings are taken mid-channel at two thirds depth, upstream of the observer, with the probe allowed to settle for sixty seconds before the value is recorded. Where flow is too shallow for the probe, a grab sample is taken and measured on the bank within ten minutes. Every instrument is checked against buffer standards on the morning of a survey and again on return.

Readings

Site	Location	Date	pH	DO mg/L	Cond. uS/cm
AC-01	Upper meadow	14 Nov	7.2	10.1	138
AC-02	Culvert outflow	14 Nov	7.0	9.6	121
AC-03	Beaver pond	15 Nov	7.3	7.4	119
AC-04	Willow bend	15 Nov	7.2	9.9	130
AC-05	Lower gauge	16 Nov	7.1	10.3	147

Notes on the culvert outflow

Conductivity at AC-02 was 121 microsiemens, which is the lowest of the five and in line with the two surveys before it. Nothing in this survey suggested the site needed watching; that came later.

Observations by site

AC-01 — Upper meadow

Visited 14 Nov. pH 7.2, dissolved oxygen 10.1 mg/L, conductivity 138 microsiemens per centimetre. Channel clear, no odour, no visible film. Bank vegetation intact and the gauge board legible from the near bank.

AC-02 — Culvert outflow

Visited 14 Nov. pH 7.0, dissolved oxygen 9.6 mg/L, conductivity 121 microsiemens per centimetre. Channel clear, no odour, no visible film. Bank vegetation intact and the gauge board legible from the near bank.

AC-03 — Beaver pond

Visited 15 Nov. pH 7.3, dissolved oxygen 7.4 mg/L, conductivity 119 microsiemens per centimetre. Channel clear, no odour, no visible film. Bank vegetation intact and the gauge board legible from the near bank.

AC-04 — Willow bend

Visited 15 Nov. pH 7.2, dissolved oxygen 9.9 mg/L, conductivity 130 microsiemens per centimetre. Channel clear, no odour, no visible film. Bank vegetation intact and the gauge board legible from the near bank.

AC-05 — Lower gauge

Visited 16 Nov. pH 7.1, dissolved oxygen 10.3 mg/L, conductivity 147 microsiemens per centimetre. Channel clear, no odour, no visible film. Bank vegetation intact and the gauge board legible from the near bank.

Recommendations

Continue the four-weekly cycle through the summer. Add a second reading above the culvert so the rise at AC-02 can be attributed rather than inferred. Replace the gauge board at AC-05, which is legible but weathered.